Large Language Models in

# Legal Reasoning & Case Outcome Prediction

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# Motivation

### High-Stakes Domain

Hallucinations or biased outputs in legal contexts can directly harm individuals. Reliability is paramount.

### Test True Reasoning

Does LLM fluency translate to structured legal reasoning — or is it surface-level pattern matching?

### AI Governance

Understanding failure modes in legal AI contributes to trustworthy AI research and responsible deployment.

**Task: Extract facts → Apply doctrine → Produce structured reasoning → Predict outcome**

# Related Work

## Legal Language Models & Benchmarks

LEGAL-BERT: domain pretraining → gains on classification (Chalkidis et al.) · LexGLUE: 7-dataset benchmark — all classification, no open reasoning · LegalBench (162 tasks): GPT-4 struggles most with interpretation tasks · MMLU: few-shot models near chance on bar-exam questions

## Legal Judgment Prediction

SVMs on ECHR data → 79% accuracy (Aletras et al.) — no reasoning trace · JEC-QA: reading-comprehension models fail on multi-hop statutory reasoning · CUAD: fine-tuned Transformers degrade on rare/complex contract clauses · ContractNLI: contractual NLI — relational reasoning at sentence level

## Structured Reasoning & Reliability

Chain-of-thought prompting (Wei et al.): boosts arithmetic & symbolic tasks · Zero-shot CoT — Lets think step by step (Kojima et al.) · RAG (Lewis et al.): ground outputs in retrieved statutes/precedents · TruthfulQA: larger models can be more confidently wrong (Lin et al.)

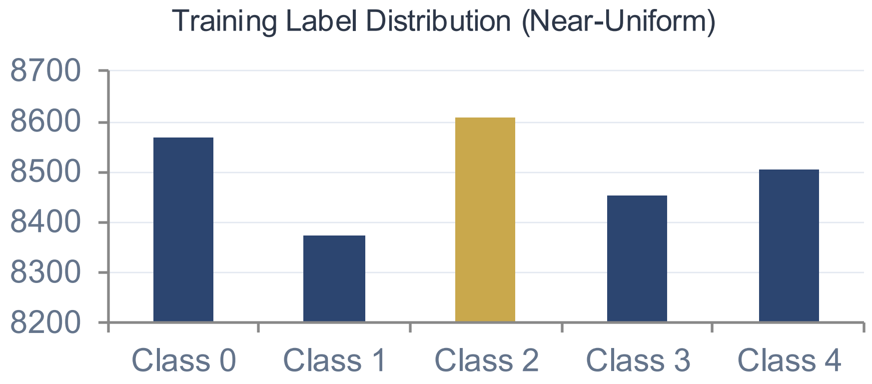
# Dataset: CaseHOLD

**53,137**
Total instances

**42,509**
Training set

**5,314**
Validation set

**5,314**
Test set




## Task & Challenges

**Task:** Given a citing context, select the correct legal holding from 5 candidates · **Domain:** US federal appellate cases, 1965–present

Subtle doctrinal distinctions · No reasoning annotations · Correct label only · Long excerpts expensive · Embedded social biases from historical cases

# Sample: What the Model Must Solve

**Citing Prompt**

*"Felony offenses that involve explosives qualify as 'violent crimes' for purposes of enhancing the sentences of career offenders. See 18 U.S.C. § 924(e)(2)(B)(ii)... Courts have found possession of a bomb to be a crime of violence... See United States v. Newman, 125 F.3d 863 (10th Cir. 1997) [HOLDING]"*

**A ✓** Holding that possession of a pipe bomb is a crime of violence for purposes of 18 USC 3142(f)(1)

**B** Holding that bank robbery by force and violence under 18 USC 2113(a) is a crime of violence

**C** Holding that sexual assault of a child qualified as crime of violence under 18 USC 16

**D** Holding for purposes of 18 USC 924(e) that being a felon in possession of a firearm is NOT a violent felony

**E** Holding that a court must only look to the statutory definition to determine if an offense is a crime of violence

# Approach & Models

## BASELINE: Majority Class Predictor

**Strategy**

Always predict class 2 (most frequent)

**Implementation**

scikit-learn DummyClassifier
strategy='most_frequent'

**Expected Result**

~20% (random chance for 5-class uniform task)

## PRIMARY: ELECTRA-small Transformer

**Architecture:** google/electra-small-discriminator + MultipleChoice head

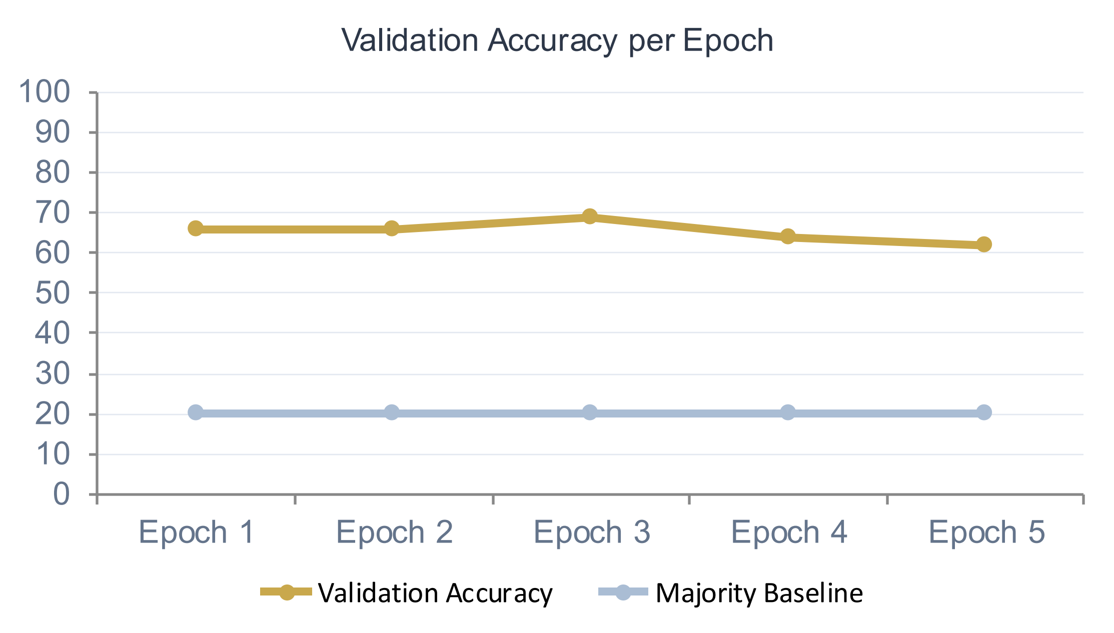
**Input:** 5 × (citing context, candidate holding) pairs → 5 logits

**Training:** 5 epochs · AdamW · lr=$2\times10^{-5}$ · batch 4

**Scale:** 1,000 train / 300 val (computational constraints)

*Planned extension (not completed due to cost/time): GPT-4 & Llama 3 with zero-shot IRAC-structured prompting*

# Results

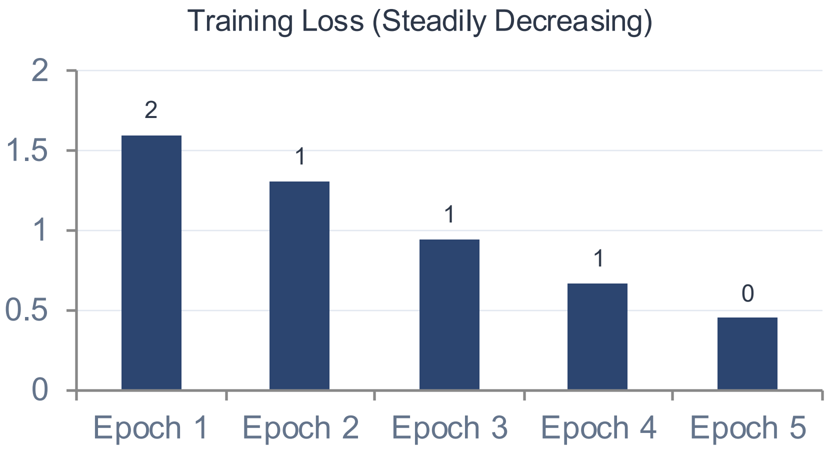



| Model | Train Size | Accuracy |
|---|---|---|
| Majority Class (ours) | – | 20.25% |
| ELECTRA-small (ours) | 1,000 | **69.00%** |
| BERT-base [prior] | ~42K | ~66% |
| Legal-BERT [prior] | ~42K | ~75% |

**3.4×** improvement over random baseline

*ELECTRA-small uses only 1.2% of available training data yet approaches BERT-base trained on full set*

# Analysis




### Genuine Predictive Signal

Legal text contains real features for holding identification which even ELECTRA-small exploits effectively

### Overfitting by Epoch 4

Loss continues falling, but val. accuracy drops 64–62%. Classic sign of memorization with only 1,000 training instances.

### Data Scale is the Bottleneck

300-instance val set = 1% accuracy per example. Full 42K training set is the clearest path to improvement.

# Conclusion & Future Work

## Key Takeaways

CaseHOLD is non-trivial; majority-class predictor confirms genuine language understanding is required

Pre-trained Transformer representations capture legally meaningful features with minimal data

Small training sample (1,000) creates overfitting; full 42K training set is the primary next step

## Future Directions

1. Train on full 42K dataset to reduce overfitting and close gap with Legal-BERT

2. Replace ELECTRA-small with Legal-BERT or DeBERTa for domain-adapted encoding

3. Evaluate GPT-4 & Llama 3 with IRAC-structured zero-shot prompting

4. Measure hallucination rate, citation accuracy, and doctrinal coherence

5. Disaggregate results by circuit, legal domain, and case era to surface bias